Chapter—5, §5C

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5C Eigenspaces and Diagonal Matrices

Definition 5.34

(Diagonal matrix)
A diagonal matrix is a square matrix
that is 0 everywhere, except possibly along the diagonal.

E.g.

$$\begin{pmatrix} 8 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Story:

- Obviously,
  every diagonal matrix is upper triangular.

- In general
  a diagonal matrix has many more 0s
  than an upper-triangular matrix.

- If an operator has
  a diagonal matrix with respect to some basis
  then
  the entries along th ediagonal are
  precisely the eigenvalues of the operator
  (this follows from Proposition 5.32)
  (there is likely an easier proof for diagonal matrices—isn't it? :)

Definition 5.36

(eigenspace, $E(\lambda, T)$)

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \boldsymbol{F}$.
The eigenspace of T corresponding to λ
denoted by $E(\lambda, T)$, is defined as
$E(\lambda, T) = \operatorname{null}(T - \lambda I)$.

In other words
$E(\lambda, T)$ is the set of all eigenvectors of T
corresponding to λ
along with the 0 vector.

NB.

- For $T \in \mathcal{L}(V)$ and $\lambda \in F$
  the eigenspace $E(\lambda, T)$ is a subspace of V
  (this is simply because the null space of each linear map on V
  is a subspace of V).

- The definitions imply that λ is an eigenvalue of T
  iff
  $E(\lambda, T) \neq \{0\}$.

Example 5.37

Suppose the matrix of an operator $T \in \mathcal{L}(V)$
with respect to a basis v_1, v_2, v_3 of V
is the matrix in Example 5.35 above.

$$\begin{pmatrix} 8 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Then
$E(8, T) = \operatorname{span}(v_1)$, $E(5, T) = \operatorname{span}(v_2, v_3)$.

Story:

- If λ is an eigenvalue of an operator $T \in \mathcal{L}(V)$
  then T restricted to $E(\lambda, T)$ is
  just the operator of multiplication by λ.

Proposition 5.38

(Sum of eigenspaces is a direct sum)

Suppose
V is finite-dimensional and
$T \in \mathcal{L}(V)$.

Suppose also that
$\lambda_1, \dots \lambda_m$ are distinct eigenvalues of T.

Then
$E(\lambda_1, T) + \cdots + E(\lambda_m, T)$

is a direct sum.

Furthermore,
$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim V$.

Proof.

To show that $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is a direct sum
suppose
$u_1 + \cdots + u_m = 0$,
where
each $u_j \in E(\lambda, T)$.

5.10 Linearly independent eigenvectors

Let $T \in \mathcal{L}(V)$. Suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of T and $v_1, \dots, v_m$ are corresponding eigenvectors. Then $v_1, \dots, v_m$ is linearly independent.

Recall Proposition 5.10
which said eigenvectors
(for distinct eigenvalues)
are linearly independent.
We therefore conclude that
each u_j must be zero.

1.44 Condition for a direct sum

Suppose $U_1, \dots, U_m$ are subspaces of V. Then $U_1 + \cdots + U_m$ is a direct sum if and only if the only way to write 0 as a sum $u_1 + \cdots + u_m$, where each u_j is in U_j, is by taking each u_j equal to 0.

Proof First suppose $U_1 + \cdots + U_m$ is a direct sum. Then the definition of direct sum implies that the only way to write 0 as a sum $u_1 + \cdots + u_m$, where each u_j is in U_j, is by taking each u_j equal to 0.

Now suppose that the only way to write 0 as a sum $u_1 + \cdots + u_m$, where each u_j is in U_j, is by taking each u_j equal to 0. To show that $U_1 + \cdots + U_m$ is a direct sum, let $v \in U_1 + \cdots + U_m$. We can write

$$v = u_1 + \cdots + u_m$$

for some $u_1 \in U_1, \dots, u_m \in U_m$. To show that this representation is unique, suppose we also have

$$v = v_1 + \cdots + v_m,$$

where $v_1 \in U_1, \dots, v_m \in U_m$. Subtracting these two equations, we have

$$0 = (u_1 - v_1) + \cdots + (u_m - v_m).$$

Because $u_1 - v_1 \in U_1, \dots, u_m - v_m \in U_m$, the equation above implies that each $u_j - v_j$ equals 0. Thus $u_1 = v_1, \dots, u_m = v_m$, as desired. ■

Using Proposition 1.44
(which says that if
$\sum_i u_i = 0$ can only be written as 0
when $u_i = 0$
then
U_is (i.e. $E(\lambda_i, T)$ here)
form a direct sum)

we conclude that $E(\lambda_i, T)$ form a direct sum.

To show the dimension part of the statement,
note that
$\dim E(\lambda_1, T) + \cdots \dim E(\lambda_m, T) = \dim \big(E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)\big) \leq \dim V$.

■

16 Suppose $U_1, \dots, U_m$ are finite-dimensional subspaces of V such that $U_1 + \cdots + U_m$ is a direct sum. Prove that $U_1 \oplus \cdots \oplus U_m$ is finite-dimensional and

$$\dim U_1 \oplus \cdots \oplus U_m = \dim U_1 + \cdots + \dim U_m.$$

SOLUTION: Suppose that $U_1, \dots, U_m$ are subspaces of V such that $V = U_1 \oplus \cdots \oplus U_m$. For each $j = 1, \dots m$, choose a basis for U_j. Put these bases together to form a single list B of vectors in V. Clearly B spans $U_1 + \cdots + U_m$, which equals V. If we show that B is also linearly independent, then it will be a basis of V. Thus the dimension of V will equal the number of vectors B. In other words, we will have

$$\dim V = \dim U_1 + \cdots + \dim U_m,$$

as desired.

We still need to show that B is linearly independent. To do this, suppose that some linear combination of B equals 0. Write this linear combination as $u_1 + \cdots + u_m$, where we have grouped together the terms that come from the basis vectors of U_1 and called their sum u_1, and similarly up to u_m. Thus we have

$$u_1 + \cdots + u_m = 0,$$

where each $u_j \in U_j$. Because $V = U_1 \oplus \cdots \oplus U_m$, this implies that each u_j equals 0. Because each u_j is a linear combination of our basis of U_j, all the

Definition 5.39

(diagonalisable)

An operator $T \in \mathcal{L}(V)$ is called diagonalisable if
the operator has a diagonal matrix
with respect to some basis of V.

Example 5.40

Define $T \in \mathcal{L}(R^2)$ by
$T(x, y) = (41x + 7y, -20x + 74y)$.

The matrix of T with respect to the standard basis of R^2 is

$$\begin{pmatrix} 41 & 7 \\ -20 & 74 \end{pmatrix},$$

which is not a diagonal matrix.

However, T is diagonalisable because
the matrix of T with respect to the basis
$(1,4), (7,5)$ is

$$\begin{pmatrix} 69 & 0 \\ 0 & 46 \end{pmatrix},$$

as one can verify.
(e.g. write the action of T on these vectors).

Proposition 5.41 (Conditions equivalent to diagonalisability)

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$.
Let $\lambda_1, \dots \lambda_m$ denote the distinct eigenvalues of T.

Then the following are equivalent

(a) T is diagonalisable
(b) V has a basis consisting of eigenvectors of T
(c) there exist 1-dimensional subspaces $U_1 \dots U_n$ of V
each invariant under T such that
$V = U_1 \oplus \cdots \oplus U_n$
(d) $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$
(e) $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

Proof.
"(a) ⇔ (b)"
An operator $T \in \mathcal{L}(V)$ has a diagonal matrix

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

with respect to a basis $v_1, \dots v_n$ of V
iff
$Tv_j = \lambda_j v_j$ for each j.

Thus, (a) and (b) are equivalent.

"(b) ⇒ (c)"
Suppose (b) holds;
thus V has a basis $v_1, \dots v_n$
consisting of eigenvectors of T.

For each j
let $U_j = \operatorname{span}(v_j)$.

Clearly,
each U_j is a 1-dimensional subspace of V
that is invariant under T.

Since
$v_1, \dots, v_n$ is a basis of V
each vector in V
can be written uniquely as
a linear combination of $v_1, \dots v_n$.
This means
each vector in V can be written uniquely
as a sum $u_1 + \cdots + u_n$
where each u_j is in U_j.

Thus
$V = U_1 \oplus \cdots \oplus U_n$.

"(c) ⇒ (b)"
Suppose (c) holds
i.e. there are 1-dimensional subspaces
$U_1, \dots U_n$ of V
each invariant under T
such that $V = U_1 \oplus \cdots \oplus U_n$.
For each j
let v_j be a nonzero vector in U_j.

Then
each v_j is an eigenvector of T.

Since each vector in V
can be written uniquely as a sum
$u_1 + \cdots + u_n$
where each u_j is in U_j
(so each u_j is a scalar multiple of v_j)
we see that $v_1, \dots v_n$ is a basis of V.

Thus (c) implies (b).

"We have (a) ⇔ (b) ⇔ (c).
We will now show that
(b) ⇒ (d) ⇒ (e) ⇒ (b)"

Recall:

(b) V has a basis consisting of eigenvectors of T;

(c) there exist 1-dimensional subspaces $U_1, \dots, U_n$ of V, each invariant under T, such that

$$V = U_1 \oplus \cdots \oplus U_n;$$

(d) $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$;

(e) $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

"(b) ⇒ (d)"
Suppose (b) holds.
Thus
V has a basis consisting of eigenvectors of T.
Hence,
every vector in V
is a linear combination of eigenvectors of T
which implies that
$V = E(\lambda_1, T) + \cdots + E(\lambda_m, T)$

and using Proposition 5.38, we get that (d) holds
i.e. $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$
(Me: one should also be able to argue directly
that the linear combination is unique
because the eigenvectors form a basis)

16 Suppose $U_1, \dots, U_m$ are finite-dimensional subspaces of V such that $U_1 + \cdots + U_m$ is a direct sum. Prove that $U_1 \oplus \cdots \oplus U_m$ is finite-dimensional and

$$\dim U_1 \oplus \cdots \oplus U_m = \dim U_1 + \cdots + \dim U_m.$$

"(d) ⇒ (e)"
We used this in the proof above as well

"(e) ⇒ (b)"
Suppose (e) holds, i.e.
$\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$. [5.42]

Choose a basis of each $E(\lambda_j, T)$
put all these bases together to form a list
$v_1, \dots v_n$ of eigenvectors of T
where $n = \dim V$
(by [5.42] above).

To show that this list is linearly independent
suppose that
$a_1 v_1 + \cdots + a_n v_n = 0$
where $a_1, \dots a_n \in F$.

For each $j \in \{1 \dots m\}$
let u_j denote the sum of all terms
$a_k v_k$ where $v_k \in E(\lambda_j, T)$.

Thus,

each u_j is in $E(\lambda_j, T)$ and
$u_1 + \cdots + u_m = 0$.

Because eigenvectors corresponding to
distinct eigenvalues are linearly independent
(see Proposition 5.10)
this implies that each u_j equals 0.

Now, since each u_j is a sum of terms
$a_k v_k$ where v_k's were chosen to be a basis
$E(\lambda_j, T)$
this implies that
all a_k's equal 0.

Thus, $v_1, \ldots v_n$ is linearly independent and
hence
is a basis of V (e.g. by Prop 2.39).

This shows (e) implies (b).

2.39 Linearly independent list of the right length is a basis

Suppose V is finite-dimensional. Then every linearly independent list of vectors in V with length $\dim V$ is a basis of V.

Proof Suppose $\dim V = n$ and $v_1, \ldots, v_n$ is linearly independent in V. The list $v_1, \ldots, v_n$ can be extended to a basis of V (by 2.33). However, every basis of V has length n, so in this case the extension is the trivial one, meaning that no elements are adjoined to $v_1, \ldots, v_n$. In other words, $v_1, \ldots, v_n$ is a basis of V, as desired. ■

■

Story:
- Not every operator is diagonalisable.
- This can happen
  even on complex vector spaces.

Example 5.43
Show that the operator $T \in \mathcal{L}(C^2)$ defined by
$T(w, z) = (z, 0)$
is not diagonalisable.

Solution.
We already saw that
for this operator,
0 is the only eigenvalue.
Further
$E(0, T) = \{(w, 0) \in C^2 : w \in C\}$.

(b), (c), (d) and (e) of Proposition 5.41
can be seen to fail on this example
(these conditions are equivalent
so it suffices to show any one fails)
Thus,
condition (a) of Proposition 5.41 also fails,
i.e. we conclude T cannot be diagonalised.

Story:
- The next result shows that
  if an operator has
  as many distinct eigenvalues
  as the dimension of its domain
  then the operator is diagonalisable

Proposition 5.44
(enough eigenvalues implies diagonalisability)

If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues
then T is diagonalisable.

Proof.
Suppose $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues
$\lambda_1, \ldots \lambda_{\dim V}$.

For each j
let $u_j \in V$ be an eigenvector corresponding to the

let $v_j \in V$ be an eigenvector corresponding to the eigenvalue λ_j.

Since eigenvectors corresponding to distinct eigenvalues are linearly independent (Proposition 5.10) $v_1, \dots, v_{\dim V}$ are linearly indpendent.

A linearly independent list of $\dim V$ vectors in V is a basis of V (e.g. Prop 2.39) thus $v_1, \dots v_{\dim V}$ is a basis of V.

Relative to this basis (consisiting of eigenvectors) T has a diagonal matrix.

■

2.39 Linearly independent list of the right length is a basis

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Proof Suppose $\dim V = n$ and $v_1, \dots, v_n$ is linearly independent in V. The list $v_1, \dots, v_n$ can be extended to a basis of V (by 2.33). However, every basis of V has length n, so in this case the extension is the trivial one, meaning that no elements are adjoined to $v_1, \dots, v_n$. In other words, $v_1, \dots, v_n$ is a basis of V, as desired. ■

Example 5.45

Define $T \in \mathcal{L}(F^3)$ by

$$T(x, y, z) = (2x + y, 5y + 3z, 8z).$$

Find a basis of F^3 with respect to which T has a diagonal matrix.

Solution With respect to the standard basis, the matrix of T is

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{pmatrix}.$$

The matrix above is upper triangular. Thus by 5.32, the eigenvalues of T are 2, 5, and 8. Because T is an operator on a vector space with dimension 3 and T has three distinct eigenvalues, 5.44 assures us that there exists a basis of $\mathbf{F}^3$ with respect to which T has a diagonal matrix.

To find this basis, we only have to find an eigenvector for each eigenvalue. In other words, we have to find a nonzero solution to the equation

$$T(x, y, z) = \lambda(x, y, z)$$

for $\lambda = 2$, then for $\lambda = 5$, and then for $\lambda = 8$. These simple equations are easy to solve: for $\lambda = 2$ we have the eigenvector $(1, 0, 0)$; for $\lambda = 5$ we have the eigenvector $(1, 3, 0)$; for $\lambda = 8$ we have the eigenvector $(1, 6, 6)$.

Thus $(1, 0, 0), (1, 3, 0), (1, 6, 6)$ is a basis of $\mathbf{F}^3$, and with respect to this basis the matrix of T is

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

The converse of 5.44 is not true. For example, the operator T defined on the three-dimensional space $\mathbf{F}^3$ by

$$T(z_1, z_2, z_3) = (4z_1, 4z_2, 5z_3)$$

has only two eigenvalues (4 and 5), but this operator has a diagonal matrix with respect to the standard basis.

In later chapters we will find additional conditions that imply that certain operators are diagonalizable.